

PHYS 598: Physics of Living Systems

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2. Intracellular signal transduction

From ligand-bound receptor to kinetic proofreading

Our discussion of the extracellular environment ended with a receptor that sensed an uncertain, and possibly ambiguous, ligand. We now ask how a cell relays that signal, filters it in time, and rejects brief or incorrect binding events. We will use dynamical systems, enzyme kinetics, and Markov processes to describe these steps.

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1 From a sensed ligand to DNA

Our discussion of the extracellular environment ended with receptor occupancy and binding-event durations. Let

$$u(t) = \text{the receptor-generated input available to the cell interior.} \quad (1)$$

Depending on the situation, u might be the fraction of occupied receptors, the number of active receptors, or an upstream activity proportional to an estimated concentration. For now, we ignore those details. We ask what the molecules inside the cell can do with $u(t)$:

$$u(t) \longrightarrow \text{what dynamical operations can molecular machinery perform?}$$

We first define the input, the variables that change over time, the output, and the timescale. We can add molecular details after we understand this simpler description.

1.1 Minimal biology: phosphorylation and dephosphorylation

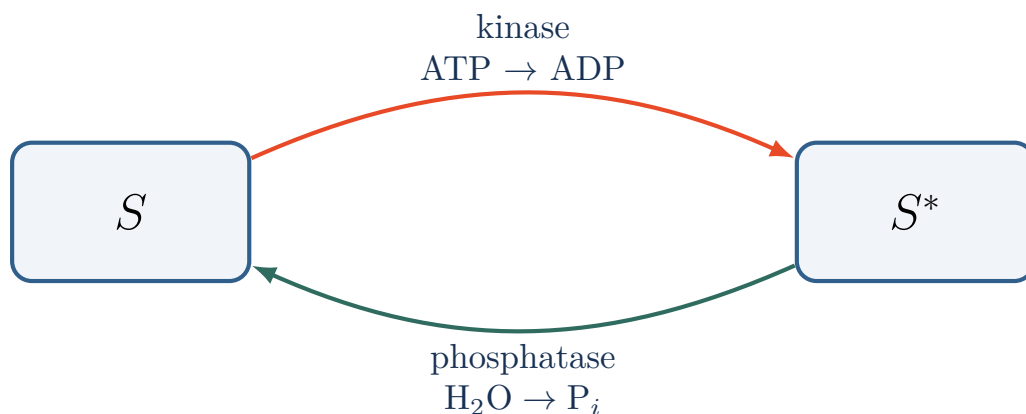
A kinase is a type of molecular machine – a molecular machine is either one protein or many proteins assembled together into one unit – that transfers a phosphate group, usually from ATP (Adenosine triphosphate), to a protein substrate:



A phosphatase is another type of molecular machine. It catalyzes hydrolysis of the phosphate ester:



The net cycle consumes chemical free energy by taking one phosphate out of ATP (called “ATP hydrolysis”). The star means “the form we call active” in our model. Real phosphorylation can activate or inhibit a protein. The notation is not a biological law [1].



active is a modeling label
phosphorylation may activate or inhibit

Figure 1: Simple re-organizing of molecules as means to activate and deactivate a molecular state. Kinase and phosphatase oppose one another, so the active fraction has an on-rate, an off-rate, and a memory time.

Note the following three points:

- The kinase is a catalyst – it is an enzyme – and is thus not consumed during the transfer of phosphate from ATP to substrate S.
- One phosphate is not usually passed from protein to protein. Every new phosphorylation event generally uses another ATP.
- The phosphatase controls the response duration and resets the system.

Why can phosphorylation act as an intracellular state variable? An occupied receptor is an event at the cell surface. By contrast, a phosphorylated protein remains in a chemically distinct state until another reaction removes the phosphate. The cell can therefore store the effect of receptor activity for a finite time in the number, or fraction, of proteins that are phosphorylated. The kinase writes this state, while the phosphatase erases it. A variable such as $x(t)$ is a coarse-grained description of this physical population of modified molecules. One often called kinase a “writer” and a phosphatase an “eraser”.

2 One activation–deactivation stage

Suppose we have a fixed number of one particular protein which can exist in either active or inactive form:

$$X_T = X + X^*, \quad (4)$$

where X and X^* denote the inactive and active forms. Suppose an inactive molecule is activated at rate $au(t)$ and an active molecule is deactivated at rate b . The activation and deactivation fluxes are therefore

$$J_+ = au(t)X, \quad J_- = bX^*. \quad (5)$$

Thus,

$$\begin{aligned} \frac{dX^*}{dt} &= J_+ - J_- \\ &= au(t)(X_T - X^*) - bX^*. \end{aligned} \quad (6)$$

It is convenient to divide by the fixed pool size and define the active fraction

$$x(t) = \frac{X^*(t)}{X_T}. \quad (7)$$

Because $X/X_T = 1 - x$, we obtain the minimal mass-action model

$$\boxed{\frac{dx}{dt} = au(t)(1 - x) - bx.} \quad (8)$$

Here au is the activation rate per inactive molecule and b is the deactivation rate per active molecule. Both have units of inverse time. If u is dimensionless, a itself has units of inverse time. If u is a concentration, a has units of inverse concentration per time.

Note that this equation does not allow x to become smaller than zero or larger than one, assuming $a, b, u(t) \geq 0$. At the lower boundary,

$$\dot{x}|_{x=0} = au(t) \geq 0, \quad (9)$$

so the dynamics cannot push x below zero. At the upper boundary,

$$\dot{x}|_{x=1} = -b \leq 0, \quad (10)$$

so the dynamics cannot push x above one. Thus, the factors $1 - x$ and x appear because the total molecular pool is finite.

Here, we assume a fixed total pool, well-mixed molecules, one effective active state, and no explicit enzyme–substrate complexes. We will later relax some of these assumptions.

Response to a constant input

For a constant step $u(t) = u_0$ beginning at $t = 0$,

$$\begin{aligned}\dot{x} &= au_0(1 - x) - bx \\ &= au_0 - (au_0 + b)x.\end{aligned}\tag{11}$$

The steady state is the value x_∞ for which $\dot{x} = 0$:

$$0 = au_0 - (au_0 + b)x_\infty,\tag{12}$$

$$(au_0 + b)x_\infty = au_0.\tag{13}$$

Hence

$$\boxed{x_\infty = \frac{au_0}{au_0 + b}}.\tag{14}$$

The subscript ∞ emphasizes the long-time value after this step. We use a superscript star, such as x^* , for a generic steady state elsewhere. That is, this value follows from balancing the opposing fluxes:

$$au_0(1 - x_\infty) = bx_\infty.\tag{15}$$

To solve the dynamics, notice from the steady-state equation that

$$au_0 = (au_0 + b)x_\infty.\tag{16}$$

Use this identity in [equation \(11\)](#):

$$\begin{aligned}\dot{x} &= (au_0 + b)x_\infty - (au_0 + b)x \\ &= -(au_0 + b)(x - x_\infty).\end{aligned}\tag{17}$$

Define the displacement from steady state,

$$y(t) = x(t) - x_\infty.\tag{18}$$

Because x_∞ is constant after the step,

$$\dot{y} = -(au_0 + b)y.\tag{19}$$

Separate variables and integrate from time zero to time t :

$$\frac{dy}{y} = -(au_0 + b) dt,\tag{20}$$

$$\int_{y(0)}^{y(t)} \frac{dy'}{y'} = -(au_0 + b) \int_0^t dt',\tag{21}$$

$$\ln \left| \frac{y(t)}{y(0)} \right| = -(au_0 + b)t.\tag{22}$$

Exponentiating gives

$$y(t) = y(0)e^{-(au_0 + b)t}.\tag{23}$$

Therefore, for an arbitrary initial active fraction,

$$x(t) = x_\infty + [x(0) - x_\infty]e^{-t/\tau}, \quad \tau = \frac{1}{au_0 + b}. \quad (24)$$

For $x(0) = 0$, this becomes

$$x(t) = x_\infty \left(1 - e^{-t/\tau}\right). \quad (25)$$

The two quantities answer different questions:

$$\begin{aligned} x_\infty(u_0) &= \text{the eventual active fraction,} \\ \tau(u_0) &= \text{the relaxation, or e-folding, time.} \end{aligned}$$

After one relaxation time, the remaining distance from steady state is reduced by e^{-1} . That is, a fraction $1 - e^{-1} \simeq 0.632$ of the total change has occurred. Both activation and deactivation restore a deviation from the new steady state, which is why their rates appear as the sum $au_0 + b$.

If the input is turned off at time t_0 , then $u = 0$ and

$$\dot{x} = -bx. \quad (26)$$

Thus

$$x(t) = x(t_0)e^{-b(t-t_0)}, \quad t \geq t_0. \quad (27)$$

The reset time is $1/b$, whereas the approach time under a constant nonzero input is $1/(au_0 + b)$. Stronger input makes the response faster in this simple model.

Weak and strong input limits

Let us introduce a dimensionless ratio

$$\rho = \frac{au_0}{b}. \quad (28)$$

Then

$$x_\infty = \frac{\rho}{1 + \rho}, \quad \tau = \frac{1/b}{1 + \rho}. \quad (29)$$

For weak activation, $\rho \ll 1$,

$$x_\infty \simeq \rho = \frac{a}{b}u_0, \quad \tau \simeq \frac{1}{b}. \quad (30)$$

Here we see that the dose response is approximately linear. This is because almost the entire protein pool remains available for activation. For strong activation, $\rho \gg 1$,

$$x_\infty \simeq 1 - \frac{1}{\rho} = 1 - \frac{b}{au_0}, \quad \tau \simeq \frac{1}{au_0}. \quad (31)$$

The response saturates because the active fraction cannot be larger than one.

We can now consider a logarithmic sensitivity S which help us more explicitly see the loss of responsiveness :

$$\begin{aligned} S &\equiv \frac{d \ln x_\infty}{d \ln u_0} \\ &= \frac{u_0}{x_\infty} \frac{dx_\infty}{du_0} \\ &= \frac{u_0}{x_\infty} \frac{ab}{(au_0 + b)^2} \\ &= \frac{b}{au_0 + b} = 1 - x_\infty. \end{aligned} \quad (32)$$

Consider a weak input, $\mathcal{S} \simeq 1$: a 1% change in input yields approximately a 1% change in output.

Now, consider a near-saturation situation, $\mathcal{S} \rightarrow 0$: a further percentage increase in input produces almost no percentage increase in output.

Although x is constant at steady state, the two chemical fluxes do not have to vanish. In fact, they can be equal:

$$J_+^* = J_-^* = X_T \frac{au_0b}{au_0 + b}. \quad (33)$$

Phosphorylation and dephosphorylation can therefore continue, with ATP being consumed, even though the mean active fraction is no longer changing. This represents a **nonequilibrium steady state**.

3 Signal cascade: nonlinear stages, linear response, gain, delay, and filtering

Bounded nonlinear cascade

Let us say that x_i be the active fraction in tier i , and define $x_0(t) \equiv u(t)$. A direct extension of the one-stage model we discussed above is

$$\begin{aligned} \dot{x}_1 &= \alpha_1 u(1 - x_1) - \beta_1 x_1, \\ \dot{x}_2 &= \alpha_2 x_1(1 - x_2) - \beta_2 x_2, \\ \dot{x}_3 &= \alpha_3 x_2(1 - x_3) - \beta_3 x_3. \end{aligned} \quad (34)$$

We can generalize the structure seen above like this:

$$\dot{x}_i = \alpha_i x_{i-1}(1 - x_i) - \beta_i x_i. \quad (35)$$

This model assumes a feed-forward cascade: tier $i - 1$ affects tier i , but the downstream tier does not appreciably deplete or otherwise load the upstream tier.

For a constant upstream activity z , one tier has the steady response

$$f_i(z) = \frac{\alpha_i z}{\alpha_i z + \beta_i}. \quad (36)$$

The exact steady cascade is therefore constructed recursively:

$$x_1^* = f_1(u_0), \quad x_2^* = f_2(x_1^*), \quad x_3^* = f_3(x_2^*), \quad (37)$$

or

$$x_3^* = f_3(f_2(f_1(u_0))). \quad (38)$$

This description is nonlinear and bounded. Each stage eventually saturates as its own active fraction approaches one.

Two different routes to a linear model

There are at least two routes to obtaining linear equations from a nonlinear biochemical cascade.

1. Low-activity approximation. If every active fraction remains small, then $1 - x_i \simeq 1$. Equation [equation \(35\)](#) becomes

$$\dot{x}_i \simeq \alpha_i x_{i-1} - \beta_i x_i. \quad (39)$$

This approximation describes motion near the inactive state. The variables themselves are small.

2. Local linearization about a nonzero working point. This is another method. A system can instead operate at substantial baseline activity while undergoing only small fluctuations about that baseline. Here, we can write

$$x_{i-1} = x_{i-1}^* + \delta x_{i-1}, \quad x_i = x_i^* + \delta x_i. \quad (40)$$

Substituting these expressions into [equation \(35\)](#) yields

$$\begin{aligned} \dot{x}_i &= \alpha_i(x_{i-1}^* + \delta x_{i-1})(1 - x_i^* - \delta x_i) - \beta_i(x_i^* + \delta x_i) \\ &= \underbrace{\alpha_i x_{i-1}^*(1 - x_i^*) - \beta_i x_i^*}_{= 0 \text{ at steady state}} + \alpha_i(1 - x_i^*)\delta x_{i-1} \\ &\quad - (\alpha_i x_{i-1}^* + \beta_i)\delta x_i - \alpha_i \delta x_{i-1} \delta x_i. \end{aligned} \quad (41)$$

The final term is a product of two small perturbations and is second order. Dropping it gives

$$\delta \dot{x}_i = A_i \delta x_{i-1} - B_i \delta x_i, \quad (42)$$

where

$$A_i = \alpha_i(1 - x_i^*), \quad B_i = \alpha_i x_{i-1}^* + \beta_i. \quad (43)$$

Note that x_i^* does not have to be small. We only assume that the perturbations δx_i are small. This expansion is in the *state variables*, not in time.

More generally, if $\dot{x}_i = F_i(x_{i-1}, x_i)$, the first-order Taylor expansion is

$$\delta \dot{x}_i = \left. \frac{\partial F_i}{\partial x_{i-1}} \right|_* \delta x_{i-1} + \left. \frac{\partial F_i}{\partial x_i} \right|_* \delta x_i. \quad (44)$$

We call the matrix made of these derivatives the Jacobian matrix. A locally stable tier has $\partial F_i / \partial x_i|_* < 0$.

For compact notation, we now write the linear cascade as

$$\begin{aligned} \dot{x}_1 &= a_1 u - b_1 x_1, \\ \dot{x}_2 &= a_2 x_1 - b_2 x_2, \\ \dot{x}_3 &= a_3 x_2 - b_3 x_3. \end{aligned} \quad (45)$$

These equations can denote either the low-activity variables themselves or small perturbations around a working point. In the second interpretation, the symbols u, x_i have been redefined to denote deviations, and a_i, b_i are effective local coefficients such as A_i, B_i above. They don't have to equal the microscopic constants in the full nonlinear model.

Steady gain in a signal cascade

At a constant input, set every derivative in [equation \(45\)](#) to zero:

$$0 = a_1 u_0 - b_1 x_1^*, \quad x_1^* = \frac{a_1}{b_1} u_0, \quad (46)$$

$$0 = a_2 x_1^* - b_2 x_2^*, \quad x_2^* = \frac{a_2}{b_2} x_1^*, \quad (47)$$

$$0 = a_3 x_2^* - b_3 x_3^*, \quad x_3^* = \frac{a_3}{b_3} x_2^*. \quad (48)$$

Substitute each result into the next:

$$x_2^* = u_0 \frac{a_1}{b_1} \frac{a_2}{b_2}, \quad (49)$$

$$x_3^* = u_0 \frac{a_1}{b_1} \frac{a_2}{b_2} \frac{a_3}{b_3}. \quad (50)$$

Therefore, defining the steady gain G_0 ,

$$x_3^* = G_0 u_0, \quad G_0 = \prod_{i=1}^3 \frac{a_i}{b_i}. \quad (51)$$

Thus, the total gain is the product of the gains from the individual stages. A large G_0 means that a fixed absolute change in input produces a large absolute change in output. It does *not* mean that the output has a larger percentage change than the input. Indeed, we can see the following:

$$\begin{aligned} \frac{d \ln x_3^*}{d \ln u_0} &= \frac{u_0}{x_3^*} \frac{dx_3^*}{du_0} \\ &= \frac{u_0}{G_0 u_0} G_0 \\ &= \boxed{1}. \end{aligned} \quad (52)$$

We see here that a 10% increase in input yields a 10% increase in output, regardless of how large G_0 is. In other words, linear amplification is not a switch-like dose response.

For the nonlinear cascade, we can write the following product of the logarithmic sensitivity of each stage:

$$\begin{aligned} \frac{d \ln x_3^*}{d \ln u_0} &= \frac{u_0}{x_3^*} \frac{dx_3^*}{dx_2^*} \frac{dx_2^*}{dx_1^*} \frac{dx_1^*}{du_0} \\ &= \left(\frac{u_0}{x_1^*} \frac{dx_1^*}{du_0} \right) \left(\frac{x_1^*}{x_2^*} \frac{dx_2^*}{dx_1^*} \right) \left(\frac{x_2^*}{x_3^*} \frac{dx_3^*}{dx_2^*} \right). \end{aligned} \quad (53)$$

For the simple bounded response

$$f_i(z) = \frac{\alpha_i z}{\alpha_i z + \beta_i}, \quad (54)$$

one tier has

$$\begin{aligned} \frac{d \ln f_i}{d \ln z} &= \frac{z}{f_i(z)} \frac{\alpha_i \beta_i}{(\alpha_i z + \beta_i)^2} \\ &= \frac{\beta_i}{\alpha_i z + \beta_i} \\ &= 1 - f_i(z) \leq 1. \end{aligned} \quad (55)$$

Thus, stacking these simple noncooperative bounded stages **does not** lead to **ultrasensitivity** (i.e., a small change in input leading to a very large change in output). Each stage becomes less responsive as it approaches saturation.

Fractional-state gain versus molecular-number gain

Suppose tier i contains $X_{i,T}$ total molecules. The active number and its small change are

$$N_i^* = X_{i,T} x_i^*, \quad \delta N_i^* = X_{i,T} \delta x_i^*. \quad (56)$$

The linear steady relation gives

$$\delta x_i^* = \frac{a_i}{b_i} \delta x_{i-1}^*. \quad (57)$$

Thus,

$$\begin{aligned} \frac{\delta N_i^*}{\delta N_{i-1}^*} &= \frac{X_{i,T} \delta x_i^*}{X_{i-1,T} \delta x_{i-1}^*} \\ &= \boxed{\frac{X_{i,T}}{X_{i-1,T}} \frac{a_i}{b_i}}. \end{aligned} \quad (58)$$

It depends on pool sizes as well as on the fractional-state gain a_i/b_i .

3.1 Repeated stages delay a step

Consider identical linear stages with removal rate b , zero initial conditions, and a step input u_0 . For simplicity, take the forward coefficient of every stage to be a :

$$\dot{x}_1 = au_0 - bx_1, \quad (59)$$

$$\dot{x}_n = ax_{n-1} - bx_n, \quad n \geq 2. \quad (60)$$

The steady values satisfy

$$x_n^* = \left(\frac{a}{b}\right)^n u_0. \quad (61)$$

Define normalized variables and dimensionless time,

$$y_n = \frac{x_n}{x_n^*}, \quad \theta = bt. \quad (62)$$

Using $ax_{n-1}^* = bx_n^*$, the equations become

$$\frac{dy_1}{d\theta} = 1 - y_1, \quad (63)$$

$$\frac{dy_n}{d\theta} = y_{n-1} - y_n, \quad n \geq 2. \quad (64)$$

The first-stage response is

$$y_1(\theta) = 1 - e^{-\theta}. \quad (65)$$

For the second stage,

$$\frac{dy_2}{d\theta} + y_2 = y_1. \quad (66)$$

Multiplying by the integrating factor e^θ gives

$$\begin{aligned} \frac{d}{d\theta}(e^\theta y_2) &= e^\theta y_1 \\ &= e^\theta - 1. \end{aligned} \quad (67)$$

Integrating from 0 to θ and using $y_2(0) = 0$,

$$\begin{aligned} e^\theta y_2 &= \int_0^\theta (e^s - 1) ds \\ &= e^\theta - 1 - \theta, \end{aligned} \quad (68)$$

so

$$y_2 = 1 - e^{-\theta}(1 + \theta). \quad (69)$$

Similarly,

$$\frac{d}{d\theta}(e^\theta y_3) = e^\theta y_2 = e^\theta - (1 + \theta). \quad (70)$$

Therefore

$$\begin{aligned} e^\theta y_3 &= \int_0^\theta [e^s - (1 + s)] ds \\ &= e^\theta - 1 - \theta - \frac{\theta^2}{2}, \end{aligned} \quad (71)$$

and

$$y_3 = 1 - e^{-\theta} \left(1 + \theta + \frac{\theta^2}{2} \right). \quad (72)$$

The same integrating-factor argument can be repeated. If

$$y_{n-1} = 1 - e^{-\theta} \sum_{m=0}^{n-2} \frac{\theta^m}{m!}, \quad (73)$$

then

$$\frac{d}{d\theta}(e^\theta y_n) = e^\theta - \sum_{m=0}^{n-2} \frac{\theta^m}{m!}. \quad (74)$$

Term-by-term integration gives

$$\boxed{\frac{x_n(t)}{x_n^*} = 1 - e^{-bt} \sum_{m=0}^{n-1} \frac{(bt)^m}{m!}}. \quad (75)$$

At early times, Taylor expansion gives

$$\frac{x_n(t)}{x_n^*} \simeq \frac{(bt)^n}{n!}. \quad (76)$$

The first tier begins responding linearly in time, the second quadratically, and the third cubically. This is the mathematical origin of the increasing delay. Differentiating the normalized step response gives the delay kernel

$$\frac{d}{dt} \left(\frac{x_n}{x_n^*} \right) = \frac{b(bt)^{n-1} e^{-bt}}{(n-1)!}. \quad (77)$$

It has the mathematical form of an **Erlang distribution**, with mean delay n/b and standard deviation $\sqrt{n}/b$. Note that this analogy does not require that one molecule literally traverse n independent checkpoints.

3.2 Frequency-domain view

We can consider a frequency-domain calculation, in which we ask how the linear cascade responds, after transients have decayed, to a small sinusoidal variation about a steady operating point. We start by writing

$$u(t) = \text{Re} \left(\tilde{u} e^{i\omega t} \right), \quad (78)$$

and seeking

$$x_j(t) = \text{Re} \left(\tilde{x}_j e^{i\omega t} \right). \quad (79)$$

Now, since we have

$$\frac{d}{dt} (\tilde{x}_j e^{i\omega t}) = i\omega \tilde{x}_j e^{i\omega t}, \quad (80)$$

the first stage gives

$$i\omega \tilde{x}_1 = a_1 \tilde{u} - b_1 \tilde{x}_1, \quad (81)$$

$$(b_1 + i\omega) \tilde{x}_1 = a_1 \tilde{u}, \quad (82)$$

and therefore

$$\frac{\tilde{x}_1}{\tilde{u}} = \frac{a_1}{b_1 + i\omega}. \quad (83)$$

Likewise,

$$\frac{\tilde{x}_2}{\tilde{x}_1} = \frac{a_2}{b_2 + i\omega}, \quad \frac{\tilde{x}_3}{\tilde{x}_2} = \frac{a_3}{b_3 + i\omega}. \quad (84)$$

Multiplication gives the transfer function

$$H(i\omega) \equiv \frac{\tilde{x}_3}{\tilde{u}} = \prod_{i=1}^3 \frac{a_i}{b_i + i\omega}. \quad (85)$$

Since $|b_i + i\omega| = \sqrt{b_i^2 + \omega^2}$,

$$|H(i\omega)| = \prod_{i=1}^3 \frac{a_i}{\sqrt{b_i^2 + \omega^2}}. \quad (86)$$

At zero frequency,

$$H(0) = \prod_{i=1}^3 \frac{a_i}{b_i} = G_0, \quad (87)$$

which is the steady gain derived above. Dividing by this value gives

$$\left| \frac{H(i\omega)}{H(0)} \right| = \prod_{i=1}^3 \frac{1}{\sqrt{1 + (\omega/b_i)^2}}. \quad (88)$$

For $\omega \ll b_i$, the corresponding factor is close to one and that stage can follow the input. For $\omega \gg b_i$, the factor is approximately b_i/ω and the rapidly varying component is strongly attenuated. The phase lag is also additive:

$$\arg H(i\omega) = - \sum_{i=1}^3 \arctan(\omega/b_i). \quad (89)$$

Thus a cascade both attenuates and delays sufficiently fast variations. This is a tradeoff: a fast true change is attenuated along with fast noise. The result applies to a genuinely linear system, or to perturbations that remain within the locally linear regime of a nonlinear one.

4 Origin of nonlinear rates: enzyme saturation

In the previous sections, the linear cascade treated each activation rate as proportional to its input. That approximation does not always hold because, for one, a cell has only finitely many enzyme molecules. We now derive the **Michaelis–Menten kinetics** that shows how finite enzyme abundance produces a saturating rate.

We start by considering the following reaction scheme involving enzyme whose concentration is E :



Here each capital letter denotes a concentration, and S denotes free substrate rather than substrate already sequestered in the enzyme-substrate complex C . The three mass-action fluxes are $k_1 ES$, $k_{-1} C$, and $k_{\text{cat}} C$. Thus,

$$[k_1] = \frac{1}{\text{concentration} \times \text{time}}, \quad [k_{-1}] = [k_{\text{cat}}] = \frac{1}{\text{time}}. \quad (91)$$

The enzyme is either free or in the complex, so $E_T = E + C$ is conserved and $E = E_T - C$. The complex and product therefore obey

$$\begin{aligned} \dot{C} &= \underbrace{k_1 ES}_{\text{complex formation}} - \underbrace{k_{-1} C}_{\text{unbinding}} - \underbrace{k_{\text{cat}} C}_{\text{catalysis}} \\ &= k_1(E_T - C)S - (k_{-1} + k_{\text{cat}})C, \end{aligned} \quad (92)$$

$$\dot{P} = k_{\text{cat}} C. \quad (93)$$

Why can we set $\dot{C} \simeq 0$? To see this, let us define

$$K_M = \frac{k_{-1} + k_{\text{cat}}}{k_1}. \quad (94)$$

Now, if we suppose that S changes little during the initial fast transient, then

$$\begin{aligned} \dot{C} &= k_1 E_T S - k_1 (S + K_M) C \\ &= k_1 (S + K_M) \left(\frac{E_T S}{S + K_M} - C \right). \end{aligned} \quad (95)$$

Thus C relaxes toward

$$C_{\text{qss}} = \frac{E_T S}{K_M + S} \quad (96)$$

on the timescale

$$\tau_C = \frac{1}{k_1 (S + K_M)}. \quad (97)$$

When this time is short compared with the time over which S is substantially depleted, the complex rapidly approaches C_{qss} and then tracks the slowly changing S . This is called the **quasi-steady-state approximation**. It does not say that C is exactly constant for the entire reaction. A common sufficient regime is $E_T \ll S_0 + K_M$.

Setting $\dot{C} \simeq 0$ in [equation \(92\)](#) now gives every algebraic step:

$$0 = k_1(E_T - C)S - (k_{-1} + k_{\text{cat}})C, \quad (98)$$

$$0 = k_1 E_T S - k_1 C S - (k_{-1} + k_{\text{cat}})C, \quad (99)$$

$$k_1 E_T S = (k_1 S + k_{-1} + k_{\text{cat}})C, \quad (100)$$

$$C = \frac{k_1 E_T S}{k_1 S + k_{-1} + k_{\text{cat}}}, \quad (101)$$

$$C = \frac{E_T S}{S + (k_{-1} + k_{\text{cat}})/k_1} = \frac{E_T S}{K_M + S}. \quad (102)$$

Because product is created at rate $v \equiv \dot{P} = k_{\text{cat}} C$,

$$\boxed{v = \frac{V_{\text{max}} S}{K_M + S}, \quad V_{\text{max}} = k_{\text{cat}} E_T.} \quad (103)$$

This derivation is the Briggs–Haldane route to Michaelis–Menten kinetics [2].

The ratio $C/E_T = S/(K_M + S)$ is the fraction of enzyme in the complex. Thus $S = K_M$ gives $C/E_T = 1/2$ and $v = V_{\max}/2$. Two limits will recur. If $S \ll K_M$, divide the numerator and denominator by K_M :

$$v = V_{\max} \frac{S/K_M}{1 + S/K_M} \simeq \frac{V_{\max}}{K_M} S. \quad (104)$$

The enzyme is mostly free and the rate is first order in S . If $S \gg K_M$, divide by S :

$$v = \frac{V_{\max}}{1 + K_M/S} \simeq V_{\max}. \quad (105)$$

Nearly every enzyme molecule is occupied, so the rate is limited by how quickly the finite enzyme pool can complete catalytic cycles. “Zero order” means approximately independent of S , not zero reaction rate.

5 The kinase–phosphatase cycle and zero-order ultrasensitivity

We now use the Michaelis–Menten rate twice. A kinase converts an inactive substrate S into its phosphorylated form S^* , while a phosphatase converts S^* back into S :



Let’s define these opposing fluxes:

$$J_+ = V_+ \frac{S}{K_+ + S}, \quad J_- = V_- \frac{S^*}{K_- + S^*}, \quad (107)$$

where $V_+ = k_{\text{cat},+} E_{+,T}$ and $V_- = k_{\text{cat},-} E_{-,T}$. Both V_+ and V_- have units of concentration per time. An upstream signal can regulate the amount or activity of the kinase and thereby change V_+ . Here V_- is the opposing reference rate.

Neglecting enzyme-bound substrate in this minimal model, $S_T = S + S^*$ is conserved. Activation adds to S^* and deactivation removes from it, so we have

$$\frac{dS^*}{dt} = J_+ - J_- = V_+ \frac{S}{K_+ + S} - V_- \frac{S^*}{K_- + S^*}. \quad (108)$$

A more exact substrate conservation law would also include kinase–substrate and phosphatase–substrate complexes. The approximation is best when those sequestered amounts are small compared with S_T .

To see which parameters control the response, let’s define the following dimensionless variables:

$$x = \frac{S^*}{S_T}, \quad \kappa_+ = \frac{K_+}{S_T}, \quad \kappa_- = \frac{K_-}{S_T}, \quad r = \frac{V_+}{V_-}. \quad (109)$$

All four quantities are dimensionless, and $0 \leq x \leq 1$. Because

$$S^* = S_T x, \quad S = S_T(1 - x), \quad \frac{dS^*}{dt} = S_T \frac{dx}{dt}, \quad (110)$$

substitution into [equation \(108\)](#) gives

$$\begin{aligned} S_T \dot{x} &= V_+ \frac{S_T(1 - x)}{K_+ + S_T(1 - x)} - V_- \frac{S_T x}{K_- + S_T x} \\ &= V_+ \frac{1 - x}{\kappa_+ + 1 - x} - V_- \frac{x}{\kappa_- + x}. \end{aligned} \quad (111)$$

Therefore the full dimensionless-state dynamics is

$$\dot{x} = \frac{V_-}{S_T} \left[r \frac{1-x}{\kappa_+ + 1-x} - \frac{x}{\kappa_- + x} \right]. \quad (112)$$

At steady state, the two nonzero fluxes balance: $J_+ = J_-$. Setting $\dot{x} = 0$ and dividing by V_- yields

$$r \frac{1-x}{\kappa_+ + 1-x} = \frac{x}{\kappa_- + x}. \quad (113)$$

This is a flux-balance condition, not thermodynamic equilibrium: phosphorylation and dephosphorylation can continue at equal nonzero rates (i.e., system is in non-equilibrium steady state).

We can also obtain the exact steady state. Cross-multiply [equation \(113\)](#):

$$r(1-x)(\kappa_- + x) = x(\kappa_+ + 1-x), \quad (114)$$

$$r\kappa_- + r(1-\kappa_-)x - rx^2 = (1+\kappa_+)x - x^2. \quad (115)$$

Moving everything to one side gives

$$(1-r)x^2 + [r(1-\kappa_-) - (1+\kappa_+)]x + r\kappa_- = 0. \quad (116)$$

Let

$$Q = 1 - r + \kappa_+ + r\kappa_-. \quad (117)$$

The root that lies in the physical interval $0 \leq x \leq 1$ is

$$x^* = \frac{2r\kappa_-}{Q + \sqrt{Q^2 - 4r\kappa_-(1-r)}}. \quad (118)$$

The exact root is useful for calculation or plotting, but the limits below make the physics clearer.

Unsaturated enzymes: a graded response

When both enzymes are unsaturated, $\kappa_{\pm} \gg 1$, [equation \(113\)](#) become

$$r \frac{1-x}{\kappa_+} \simeq \frac{x}{\kappa_-}, \quad (119)$$

$$r\kappa_-(1-x) = \kappa_+x, \quad (120)$$

$$r\kappa_- = (\kappa_+ + r\kappa_-)x. \quad (121)$$

Hence

$$x^* \simeq \frac{r\kappa_-}{\kappa_+ + r\kappa_-}. \quad (122)$$

For equal Michaelis constants, the above reduces to $x^* = r/(1+r)$, a broad, graded response.

Saturated enzymes: why the response becomes sharp

When $\kappa_{\pm} \ll 1$ and x is not close to zero or one, both substrate fractions in [equation \(112\)](#) are nearly one. Therefore

$$\dot{x} \simeq \frac{V_+ - V_-}{S_T}. \quad (123)$$

If $V_+ < V_-$ ($r < 1$), the pool is driven toward $x = 0$ until the phosphatase runs short of its substrate S^* and desaturates. If $V_+ > V_-$ ($r > 1$), the pool is driven toward $x = 1$ until the kinase runs short of S and desaturates. This motion toward one endpoint gives the sharp response.

For $r < 1$, anticipate $x \ll 1$. The kinase remains saturated, whereas the phosphatase must be allowed to desaturate. Then

$$r \simeq \frac{x}{\kappa_- + x}, \quad (124)$$

$$r(\kappa_- + x) = x, \quad (125)$$

$$r\kappa_- = (1 - r)x, \quad (126)$$

so

$$\boxed{x^* \simeq \frac{r\kappa_-}{1 - r}}. \quad (127)$$

This branch is self-consistent when $1 - r \gg r\kappa_-$. For $r > 1$, let $y = 1 - x \ll 1$. The phosphatase remains saturated, while the kinase desaturates:

$$r \frac{y}{\kappa_+ + y} \simeq 1, \quad (128)$$

$$ry = \kappa_+ + y, \quad (129)$$

$$(r - 1)y = \kappa_+. \quad (130)$$

Therefore

$$\boxed{1 - x^* = y \simeq \frac{\kappa_+}{r - 1}}. \quad (131)$$

This branch is self-consistent when $r - 1 \gg \kappa_+$. Both endpoint approximations fail in the narrow transition near $r = 1$, where [equation \(118\)](#) connects them. Exactly at $r = 1$,

$$x^* = \frac{\kappa_-}{\kappa_+ + \kappa_-}. \quad (132)$$

For $\kappa_+ = \kappa_- = \kappa$, this is $x^* = 1/2$.

We can quantify “sharp” in this symmetric case without using the quadratic formula. For $\kappa_+ = \kappa_- = \kappa$, solve the flux balance for r :

$$r(x) = \frac{x(\kappa + 1 - x)}{(\kappa + x)(1 - x)}. \quad (133)$$

Taking the logarithm and differentiating term by term gives

$$\frac{1}{r} \frac{dr}{dx} = \frac{1}{x} - \frac{1}{\kappa + 1 - x} - \frac{1}{\kappa + x} + \frac{1}{1 - x}. \quad (134)$$

At $r = 1$ and $x^* = 1/2$,

$$\begin{aligned} \left. \frac{dr}{dx} \right|_{x=1/2} &= 4 - \frac{2}{\kappa + 1/2} \\ &= \frac{8\kappa}{1 + 2\kappa}. \end{aligned} \quad (135)$$

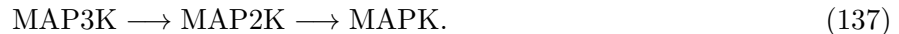
Inverting this slope gives

$$\boxed{\left. \frac{dx^*}{dr} \right|_{r=1} = \frac{1 + 2\kappa}{8\kappa}}. \quad (136)$$

The slope grows without bound as $\kappa \rightarrow 0$. Thus, when $K_{\pm} \ll S_T$, a small change in the balance of maximal kinase and phosphatase rates can move most of the conserved pool between states. This is the **zero-order ultrasensitivity**, a result due to Goldbeter and Koshland [3].

6 MAP kinase as a biological realization of signal-transduction cascade

In many living cells, a canonical three-tier signal-transduction module is



MAPK stands for **mitogen-activated protein kinase**, although members of the MAPK family respond to many signals other than mitogens. The K in each abbreviation means kinase. An active kinase at one tier catalyzes phosphorylation of molecules in the next tier, while a phosphatase opposes the reaction at every tier. The upstream kinase is a catalyst: it is not consumed, and a single phosphate is not literally passed down the chain.

We can apply the preceding mathematics separately to each tier. Let X_i^* be the active form at tier i , let $x_i = X_i^*/X_{i,T}$, and let $X_i = X_{i,T}(1 - x_i)$ be the inactive form. A coarse-grained tier obeys

$$\dot{x}_i = \frac{V_{+,i}(x_{i-1})}{X_{i,T}} \frac{1 - x_i}{\kappa_{+,i} + 1 - x_i} - \frac{V_{-,i}}{X_{i,T}} \frac{x_i}{\kappa_{-,i} + x_i}, \quad (138)$$

where $\kappa_{\pm,i} = K_{\pm,i}/X_{i,T}$. The active population in the preceding tier acts as the kinase enzyme for this tier. For example, a minimal choice is

$$V_{+,i}(x_{i-1}) = k_{\text{cat},i}^+ X_{i-1,T} x_{i-1}. \quad (139)$$

Thus the output of tier $i - 1$ controls the forward flux of tier i .

This nonlinear equation also connects directly to our earlier discussion of linear cascade. If both enzymes are unsaturated and the active fraction is small, then $\kappa_{\pm,i} \gg 1$, $1 - x_i \simeq 1$, and

$$\begin{aligned} \dot{x}_i &\simeq \frac{k_{\text{cat},i}^+ X_{i-1,T}}{X_{i,T} \kappa_{+,i}} x_{i-1} - \frac{V_{-,i}}{X_{i,T} \kappa_{-,i}} x_i \\ &= a_i x_{i-1} - b_i x_i. \end{aligned} \quad (140)$$

Thus, the linear model is the weak-activity, unsaturated limit of the nonlinear biochemical model. More generally, it is the local small-perturbation approximation. When enzymes saturate, the same tiers can instead show a steep steady-state response.

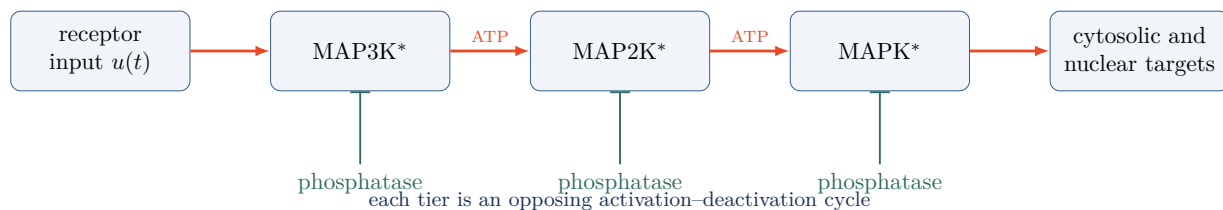


Figure 2: A coarse-grained MAPK relay. The arrows summarize enzymatic processes in which kinase adds a phosphate by taking phosphate P out of ATP. Each arrow labeled 'ATP' uses a new ATP (i.e., ATP hydrolysis occurs for every red arrow with "ATP" label).

In the budding yeast's mating pathway, receptor-activated $G\beta\gamma$ recruits the Ste5-bound Ste11–Ste7–Fus3 module to the membrane, where Ste20 activates Ste11 [4, 5]. This is only one example of MAPK signaling. Different MAPK-family pathways use different receptors, scaffolds, spatial organization, feedbacks, and outputs. Fus3 also regulates cytosolic processes.

MAPK systems may include enzyme saturation, dual phosphorylation, sequestration, spatial organization, and feedback [6, 7].

7 Kinetic proofreading: using time and energy to improve specificity

7.1 Question that prompted kinetic proofreading

During 1974 and 1975, Hopfield and, independently, Ninio proposed a mechanism, **kinetic proofreading**, for distinguishing signal triggered by a decoy molecule from signal triggered by the desired, target molecule. They were motivated by these considerations:

1. Unusually accurate biosynthetic processes were already known or estimated experimentally.
2. An ordinary one-step recognition model seemed unable to explain all of that accuracy from the available molecular differences.

After their theoretical predictions, researchers found a direct experimental evidence for proofreading in an aminoacyl-tRNA synthetase system [8]. The original papers concerned biosynthetic product selection. But in our course, we will consider a simpler model that still captures the essence of the Hopfield's and Ninio's original model: the receptor-signaling chain. This model is closely related to McKeithan's model of receptor signaling, written down in 1995 [9].

Ordinary equilibrium binding and kinetic proofreading ask different questions:

equilibrium binding asks: which ligand occupies the receptor?

proofreading asks: which bound encounter completes a driven checklist before it ends?

The cell does not have to identify the molecule by name. It can apply the same sequence to every bound ligand and let different bound lifetimes produce different probabilities of completion.

7.2 Error that we want to reduce

Let us first define what we mean by an "error". Let T be a target, or correct, ligand and let D denote a decoy, or incorrect, ligand. Suppose the two ligands produce amounts of output O_T and O_D according to some stated measurement rule. We will use the **decoy-to-target error odds**

$$\boxed{\eta \equiv \frac{O_D}{O_T}}. \quad (141)$$

Small η means that the desired output dominates. The fraction of all output that is wrong is a different, though related, quantity:

$$\begin{aligned} \epsilon &\equiv \frac{O_D}{O_T + O_D} \\ &= \frac{O_D/O_T}{1 + O_D/O_T} = \boxed{\frac{\eta}{1 + \eta}}. \end{aligned} \quad (142)$$

Conversely, $\eta = \epsilon/(1 - \epsilon)$. When $\eta \ll 1$, the two numbers are approximately equal. But they don't mean the same thing. For example, $\eta = 0.1$ means one wrong output per ten right outputs, while the wrong-output fraction is $0.1/1.1 \simeq 0.091$.

Here, O_i denotes the output produced by ligand i . A biosynthetic reaction might count completed products. A receptor might count entries into an active state or integrate signal continuously while that state remains occupied. These choices lead to closely related but not identical formulas. We will derive both and name the observable before taking a ratio.

7.3 Benchmark to compare to: one-step equilibrium

Before discussing what proofreading is, let's consider ordinary reversible binding,



Here c_i is the free concentration of ligand i , R is the amount of free receptor, and C_i is the amount of receptor bound to that ligand. Let's define

$$a_i \equiv k_{\text{on},i} c_i, \quad r_i \equiv k_{\text{off},i}, \quad \tau_i \equiv \frac{1}{r_i}. \quad (144)$$

The quantity a_i is the rate at which one free receptor is hit by ligand i . It has units of inverse time. The quantity r_i is the dissociation rate of a bound complex, and τ_i is its mean bound lifetime.

Mass action gives

$$\frac{dC_i}{dt} = a_i R - r_i C_i. \quad (145)$$

At equilibrium the binding and unbinding fluxes balance:

$$\begin{aligned} 0 &= a_i R - r_i C_i^{\text{eq}}, \\ \frac{C_i^{\text{eq}}}{R} &= \frac{a_i}{r_i} = \frac{k_{\text{on},i} c_i}{k_{\text{off},i}} = \frac{c_i}{K_{d,i}}, \end{aligned} \quad (146)$$

where $K_{d,i} = k_{\text{off},i}/k_{\text{on},i}$. Dividing the decoy result by the target result gives the equilibrium occupancy odds

$$\begin{aligned} \eta_{\text{eq}} &\equiv \frac{C_D^{\text{eq}}}{C_T^{\text{eq}}} \\ &= \frac{c_D/K_{d,D}}{c_T/K_{d,T}} = \boxed{\frac{a_D}{a_T} \frac{r_T}{r_D}}. \end{aligned} \quad (147)$$

This equation shows two ways in which a decoy can compete with the target. A decoy can be favored because it is abundant, through a_D/a_T , even if each decoy complex is less stable, through $r_T/r_D < 1$.

What does one equilibrium binding-energy difference mean? Let

$$\Delta\Delta G \equiv \Delta G_{\text{bind},D}^{\circ} - \Delta G_{\text{bind},T}^{\circ} > 0, \quad (148)$$

so target binding is more favorable. Equilibrium statistical mechanics relates the dissociation constants by

$$\frac{K_{d,T}}{K_{d,D}} = \exp\left(-\frac{\Delta\Delta G}{k_B T}\right). \quad (149)$$

Substitution into the occupancy ratio gives

$$\boxed{\eta_{\text{eq}} = \frac{c_D}{c_T} \exp\left(-\frac{\Delta\Delta G}{k_B T}\right)}. \quad (150)$$

At equal ligand concentrations, the correct–incorrect free-energy difference appears once, as one Boltzmann factor. It is not one chemical bond and it is not a claim that the receptor measures energy directly. It is the total standard binding-free-energy advantage of the target in this simple equilibrium model.

For example,

$$\Delta\Delta G = k_B T \ln 100 \simeq 4.6 k_B T \quad (151)$$

gives $\eta_{\text{eq}} = 10^{-2}$ at equal concentrations: about one wrong bound complex per one hundred right bound complexes. If a final product error near 10^{-4} is observed, a natural question is how the network obtained two orders of magnitude beyond this one-use benchmark.

7.4 How can a driven network improve on the benchmark set by one-step equilibrium?

Adding more equilibrium states does not improve the discrimination by itself. At equilibrium, detailed balance constrains the forward and reverse fluxes, and the network cannot freely use the same molecular advantage again at each intermediate step. Hopfield's key proposal was to couple intermediate transitions to a source of chemical free energy, such as nucleotide hydrolysis. Strong chemical driving can make the sequence effectively directional on the timescale of interest.

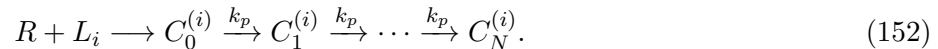
Directionality alone is also insufficient. Dissociation must erase progress. Otherwise several short encounters could accumulate the same downstream modifications as one long encounter. For proofreading to work in our model, the system needs the following four features:

1. It has one or more bound intermediate states.
2. A forward transition competes with ligand dissociation.
3. Rapid reset to the beginning occurs when the ligand dissociates.
4. Chemical driving biases the forward transitions.

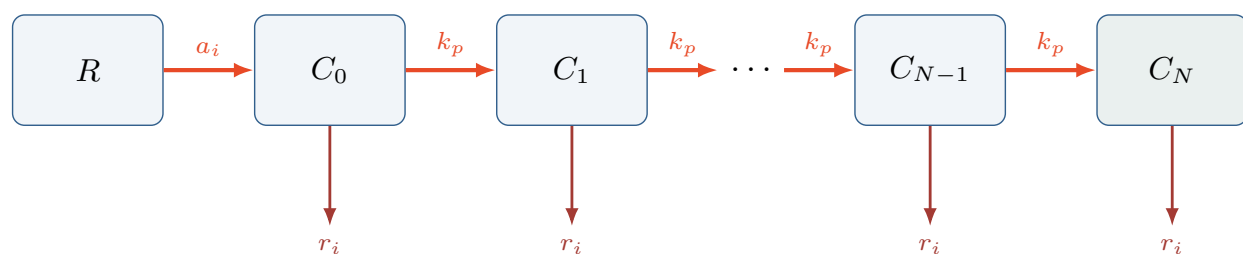
Here we note that in what follows, the irreversible arrows we draw below are approximation of strong drive (e.g. ATP hydrolysis fueling one way, making the reverse energetically highly unlikely). A thermodynamically complete model would show fuel molecules and reverse reactions explicitly.

7.5 A simplified kinetic-proofreading chain

For ligand i , consider the following signal cascade:



Here, $C_0^{(i)}$ denotes the complex immediately after binding. $N \geq 1$ is the number of forward arrows that must be completed before the terminal state $C_N^{(i)}$ is reached. Ligand can dissociate from every state at rate r_i , immediately returning the receptor to its unmodified starting state.



dissociation from every state resets progress
reaching C_N requires N successful races

Figure 3: A coarse-grained kinetic-proofreading chain. At every nonterminal state, a forward modification competes with ligand dissociation. Dissociation erases progress. The forward arrows stand for chemically driven transitions. The fuel (ATP) and energetically unfavorable reverse reactions are omitted from this simplified drawing.

Let's assume the following:

1. Ligand concentrations remain fixed reservoirs, so the attempt rate per free receptor is $a_i = k_{\text{on},i}c_i$.
2. One receptor binds at most one ligand at a time.
3. Every forward step has the same constant rate k_p for target and decoy.
4. Dissociation has constant rate r_i from every intermediate and is independent of forward progression.
5. Dissociation produces rapid, complete reset.
6. Forward reactions are treated as irreversible.
7. Receptors do not cooperate with one another.

These assumptions make the first calculation simple. Real pathways need not obey all of them.

7.6 Why a constant reaction rate gives an exponential waiting time

Here we use the notation $T \sim \text{Exp}(k)$ to mean that T is exponentially distributed with rate k . It follows from the meaning of a constant first-order rate. Let's delve a bit deeper. Suppose an event has not yet occurred by time t . A constant rate k at which the event can occur means:

$$P(\text{event in } [t, t + dt] \mid \text{no event before } t) = k \, dt + O(dt^2). \quad (153)$$

Let

$$\mathcal{S}(t) \equiv P(T > t) \quad (154)$$

be the **survival probability: the probability that the waiting time exceeds t** . To survive until $t + dt$, the system must first survive to t and then avoid the event during the next short interval. Therefore

$$\mathcal{S}(t + dt) = \mathcal{S}(t) (1 - k \, dt) + O(dt^2). \quad (155)$$

Subtract $\mathcal{S}(t)$, divide by dt , and take the limit:

$$\frac{d\mathcal{S}}{dt} = -k\mathcal{S}, \quad \mathcal{S}(0) = 1. \quad (156)$$

The solution is

$$\boxed{\mathcal{S}(t) = e^{-kt}}. \quad (157)$$

The **probability density** of the event time is the rate at which survival probability is lost:

$$f_T(t) = -\frac{d\mathcal{S}}{dt} = ke^{-kt}, \quad t \geq 0. \quad (158)$$

Its mean is

$$\langle T \rangle = \int_0^\infty tke^{-kt} \, dt = \frac{1}{k}. \quad (159)$$

Thus

$$T_p \sim \text{Exp}(k_p), \quad T_{\text{off}}^{(i)} \sim \text{Exp}(r_i) \quad (160)$$

means that progression and dissociation are being modeled as time-homogeneous, **first-order Markov events** with mean waiting times $1/k_p$ and $1/r_i$.

The exponential distribution is memoryless:

$$\begin{aligned} P(T > s + t \mid T > s) &= \frac{P(T > s + t)}{P(T > s)} \\ &= \frac{e^{-k(s+t)}}{e^{-ks}} = e^{-kt} = P(T > t). \end{aligned} \quad (161)$$

After a complex has survived one checkpoint, its *remaining* dissociation time has the same distribution as it had when that checkpoint began. This property will let us multiply checkpoint probabilities.

7.7 One checkpoint is a race between two clocks

Immediately after a complex enters any nonterminal state, two events can occur:

1. A forward modification occurs after a random waiting time T_p .
2. The ligand dissociates after a random waiting time $T_{\text{off}}^{(i)}$.

The event

$$T_p < T_{\text{off}}^{(i)} \quad (162)$$

means exactly that progression happens first. It is the event that this checkpoint is passed. The variables are mathematical representations of two possible reaction times.

Let's now calculate this probability. The probability that progression occurs between t and $t + dt$ is

$$f_{T_p}(t) dt = k_p e^{-k_p t} dt. \quad (163)$$

For this progression to count as a win, dissociation must not have occurred before t . The required survival probability is

$$P(T_{\text{off}}^{(i)} > t) = e^{-r_i t}. \quad (164)$$

Because progression and dissociation are assumed independent, the probability that progression occurs in the small interval while the ligand is still bound is the product

$$k_p e^{-k_p t} e^{-r_i t} dt. \quad (165)$$

The winning progression could occur at any $t \geq 0$, so we add these mutually exclusive possibilities by integrating:

$$\begin{aligned} p_i &\equiv P(T_p < T_{\text{off}}^{(i)}) \\ &= \int_0^\infty f_{T_p}(t) P(T_{\text{off}}^{(i)} > t) dt \\ &= \int_0^\infty k_p e^{-k_p t} e^{-r_i t} dt \\ &= k_p \int_0^\infty e^{-(k_p + r_i)t} dt \\ &= k_p \left[-\frac{e^{-(k_p + r_i)t}}{k_p + r_i} \right]_0^\infty \\ &= \boxed{\frac{k_p}{k_p + r_i}}. \end{aligned} \quad (166)$$

There is a useful shorter interpretation. The total rate for leaving the intermediate is $k_p + r_i$. The fraction of that total rate belonging to the forward channel is $k_p/(k_p + r_i)$. Similarly,

$$P(T_{\text{off}}^{(i)} < T_p) = \frac{r_i}{k_p + r_i}, \quad (167)$$

and the success and failure probabilities add to one.

Let us check three limits:

$$\begin{aligned} k_p \gg r_i &\implies p_i \simeq 1, && \text{progression usually wins,} \\ k_p \ll r_i &\implies p_i \simeq \frac{k_p}{r_i} \ll 1, && \text{dissociation usually wins,} \\ k_p = r_i &\implies p_i = \frac{1}{2}, && \text{the two outcomes are equally likely.} \end{aligned}$$

If the target is longer lived, then $r_T < r_D$ and

$$p_T = \frac{k_p}{k_p + r_T} > \frac{k_p}{k_p + r_D} = p_D. \quad (168)$$

The machinery uses the same k_p for both ligands. The difference arises because the shorter-lived decoy loses more races to dissociation.

7.8 From one checkpoint to N checkpoints: why probabilities multiply

Let A_n be the event that checkpoint n is passed before the ligand dissociates. For two checkpoints, the probability of completion is

$$\begin{aligned} P(A_1 \text{ and } A_2) &= P(A_1)P(A_2 | A_1) \\ &= p_i p_i = p_i^2. \end{aligned} \quad (169)$$

The second factor equals p_i because, conditional on survival through the first step, the remaining ligand lifetime is again exponential with rate r_i by [equation \(161\)](#). The new forward step also has rate k_p . Thus the same race begins in the new state.

For N steps, the multiplication rule gives

$$\begin{aligned} q_i &\equiv P(A_1 \text{ and } A_2 \text{ and } \dots \text{ and } A_N) \\ &= P(A_1)P(A_2 | A_1) \dots P(A_N | A_1, \dots, A_{N-1}) \\ &= p_i^N. \end{aligned} \quad (170)$$

Alternatively, let $q_n^{(i)}$ be the probability of eventually reaching $C_N^{(i)}$ before dissociation when the complex currently occupies $C_n^{(i)}$. At the end,

$$q_N^{(i)} = 1, \quad (171)$$

because a complex already in the terminal state has completed the checklist. For $n < N$, condition on the next event:

$$\begin{aligned} q_n^{(i)} &= P(\text{progress next})q_{n+1}^{(i)} + P(\text{dissociate next}) \times 0 \\ &= p_i q_{n+1}^{(i)}. \end{aligned} \quad (172)$$

Working backward,

$$q_{N-1}^{(i)} = p_i, \quad q_{N-2}^{(i)} = p_i^2, \quad \dots, \quad q_0^{(i)} = p_i^N. \quad (173)$$

Both arguments give

$$q_i = P(\text{complete } N \text{ steps} | \text{binding}) = p_i^N = \left(\frac{k_p}{k_p + r_i} \right)^N. \quad (174)$$

This formula describes an ensemble of binding attempts. **An individual trajectory does not complete a fractional number of checkpoints: it either reaches $C_N^{(i)}$ or dissociates first.** Nor is the chain a deterministic timer that accepts every ligand lasting longer than a fixed threshold. **The ligand must remain bound while N random forward transitions occur.** A decoy can occasionally get lucky and advance rapidly. A target can still dissociate early.

7.9 The same stochastic process as ensemble rate equations

The preceding calculation followed one binding attempt. We can describe many independent receptors by equations for their mean occupancies. Here $C_n^{(i)}$ denotes the concentration, or mean copy number, of receptors bound to ligand i in state n . The equations are

$$\begin{aligned}\dot{C}_0^{(i)} &= \underbrace{a_i R}_{\text{new binding}} - \underbrace{k_p C_0^{(i)}}_{\text{forward}} - \underbrace{r_i C_0^{(i)}}_{\text{dissociation}} \\ &= a_i R - (k_p + r_i) C_0^{(i)}, \\ \dot{C}_n^{(i)} &= \underbrace{k_p C_{n-1}^{(i)}}_{\text{forward influx}} - \underbrace{k_p C_n^{(i)}}_{\text{forward outflux}} - \underbrace{r_i C_n^{(i)}}_{\text{dissociation}}, \quad 1 \leq n < N, \\ &= k_p C_{n-1}^{(i)} - (k_p + r_i) C_n^{(i)}, \quad 1 \leq n < N, \\ \dot{C}_N^{(i)} &= \underbrace{k_p C_{N-1}^{(i)}}_{\text{terminal influx}} - \underbrace{r_i C_N^{(i)}}_{\text{dissociation}}.\end{aligned}\tag{175}$$

There is no $-k_p C_N^{(i)}$ term in the final equation because the drawn chain has no further proofreading state. The terminal complex remains active until the ligand dissociates.

If all dissociation events return a receptor to the free pool, then

$$\dot{R} = - \sum_i a_i R + \sum_i r_i \sum_{n=0}^N C_n^{(i)}.\tag{176}$$

The loss term counts binding of either ligand. The gain term counts dissociation from every bound state. Together, the equations conserve total receptor:

$$R_{\text{tot}} = R + \sum_i \sum_{n=0}^N C_n^{(i)}.\tag{177}$$

For this linear, noncooperative model, these occupancy equations are the ensemble equations of the same continuous-time Markov process used above. They do not say that an individual receptor follows a smooth deterministic trajectory.

Steady-state solution

Let's set each derivative in [equation \(175\)](#) equal to zero. Then the first state obeys

$$\begin{aligned}0 &= a_i R - (k_p + r_i) C_0^{(i)}, \\ C_0^{(i)} &= \frac{a_i R}{k_p + r_i}.\end{aligned}\tag{178}$$

For any nonterminal state $1 \leq n < N$,

$$\begin{aligned}0 &= k_p C_{n-1}^{(i)} - (k_p + r_i) C_n^{(i)}, \\ C_n^{(i)} &= \frac{k_p}{k_p + r_i} C_{n-1}^{(i)} = p_i C_{n-1}^{(i)}.\end{aligned}\tag{179}$$

Applying this relation repeatedly,

$$\begin{aligned}
 C_1^{(i)} &= p_i C_0^{(i)}, \\
 C_2^{(i)} &= p_i C_1^{(i)} = p_i^2 C_0^{(i)}, \\
 &\vdots \\
 C_{N-1}^{(i)} &= p_i^{N-1} C_0^{(i)} = \frac{a_i R}{k_p + r_i} p_i^{N-1}.
 \end{aligned} \tag{180}$$

At the terminal state, steady influx balances dissociation:

$$\begin{aligned}
 0 &= k_p C_{N-1}^{(i)} - r_i C_N^{(i)}, \\
 C_N^{(i)} &= \frac{k_p}{r_i} C_{N-1}^{(i)} \\
 &= \frac{k_p}{r_i} \frac{a_i R}{k_p + r_i} p_i^{N-1} \\
 &= \boxed{\frac{a_i R}{r_i} p_i^N}.
 \end{aligned} \tag{181}$$

The flux of complexes entering the terminal state is

$$\begin{aligned}
 J_i &\equiv k_p C_{N-1}^{(i)} \\
 &= k_p \frac{a_i R}{k_p + r_i} p_i^{N-1} \\
 &= \boxed{a_i R p_i^N = a_i R q_i}.
 \end{aligned} \tag{182}$$

This agrees with the single-attempt argument:

$$\underbrace{J_i}_{\text{successful completions per time}} = \underbrace{a_i R}_{\text{binding attempts per time}} \times \underbrace{q_i}_{\text{success probability per attempt}}.$$

Finally, compare [equations \(181\) and \(182\)](#):

$$\boxed{C_N^{(i)} = \frac{J_i}{r_i} = J_i \tau_i}. \tag{183}$$

This is the familiar relation

$$\text{mean occupancy} = \text{arrival flux} \times \text{mean residence time}.$$

The residence-time factor will explain why counting terminal entries and integrating terminal activity give different discrimination formulas.

Where did competition for free receptors go?

Target and decoy share the same free-receptor pool. To find R , first sum the nonterminal states for one ligand:

$$\begin{aligned}
 \sum_{n=0}^{N-1} C_n^{(i)} &= \frac{a_i R}{k_p + r_i} \sum_{n=0}^{N-1} p_i^n \\
 &= \frac{a_i R}{k_p + r_i} \frac{1 - p_i^N}{1 - p_i}.
 \end{aligned} \tag{184}$$

Because

$$1 - p_i = 1 - \frac{k_p}{k_p + r_i} = \frac{r_i}{k_p + r_i}, \quad (185)$$

the sum becomes

$$\sum_{n=0}^{N-1} C_n^{(i)} = \frac{a_i R}{r_i} (1 - p_i^N). \quad (186)$$

Adding the terminal occupancy gives

$$\begin{aligned} \sum_{n=0}^N C_n^{(i)} &= \frac{a_i R}{r_i} (1 - p_i^N) + \frac{a_i R}{r_i} p_i^N \\ &= \boxed{\frac{a_i R}{r_i}}. \end{aligned} \quad (187)$$

The total occupancy does not depend on N in this idealized model. Proofreading redistributes a bound receptor among internal states, because its ligand has the same dissociation rate from every state.

Substitute [equation \(187\)](#) into receptor conservation:

$$\begin{aligned} R_{\text{tot}} &= R + R \sum_i \frac{a_i}{r_i} \\ &= R \left(1 + \sum_i \frac{a_i}{r_i} \right). \end{aligned} \quad (188)$$

Therefore

$$\boxed{R = \frac{R_{\text{tot}}}{1 + \sum_i a_i / r_i}}. \quad (189)$$

The common factor R will cancel from target-to-decoy ratios. It does not follow that receptor competition is unimportant: an abundant decoy can make R small and reduce the *absolute* target completion rate.

7.10 Two definitions of the output

We have derived five important quantities :

Symbol	Units	Meaning
$p_i = k_p / (k_p + r_i)$	dimensionless	probability of passing one checkpoint
$q_i = p_i^N$	dimensionless	probability of completing all checkpoints per binding attempt
$J_i = a_i R q_i$	amount per time	flux of entries into the terminal state
$C_N^{(i)} = J_i / r_i$	amount	mean occupancy of the terminal state
$S_i = k_s C_N^{(i)}$	signal amount per time	continuous signal-production rate from terminal complexes

Let's consider two definitions of the output.

Output definition 1: one accepted product or pulse per completion. If entering $C_N^{(i)}$ produces one product or one equal-sized pulse, the output is the completion flux

$$O_i^{\text{entry}} = J_i = a_i R p_i^N. \quad (190)$$

This convention is closest to the historical question of correct and incorrect biosynthetic products.

Output definition 2: continuous signaling while the terminal state remains occupied. Suppose each terminal complex produces messenger continuously at rate k_s . Then

$$O_i^{\text{cont}} = S_i = k_s C_N^{(i)} = k_s \frac{a_i R}{r_i} p_i^N. \quad (191)$$

This convention is plausible for a receptor that remains catalytically active. If we formally set $N = 0$, the initially bound state is terminal and $C_0^{(i)} = a_i R / r_i$. Thus, the continuous convention reduces to ordinary bound-state signaling.

Which definition we use depends on what the downstream molecules measure. We will consider both below.

7.11 Target–decoy discrimination

Completed-entry or product error (output definition 1)

As output, suppose we use the output definition 1: one product or pulse per completion. Then the **decoy-to-target error odds** are

$$\begin{aligned} \eta_N^{\text{entry}} &\equiv \frac{J_D}{J_T} \\ &= \frac{a_D R p_D^N}{a_T R p_T^N} \\ &= \frac{a_D}{a_T} \left(\frac{p_D}{p_T} \right)^N. \end{aligned} \quad (192)$$

The shared free-receptor amount cancels. Now evaluate the ratio of the one-checkpoint probabilities:

$$\begin{aligned} \frac{p_D}{p_T} &= \frac{k_p / (k_p + r_D)}{k_p / (k_p + r_T)} \\ &= \frac{k_p}{k_p + r_D} \frac{k_p + r_T}{k_p} \\ &= \frac{k_p + r_T}{k_p + r_D}. \end{aligned} \quad (193)$$

Let us define the factor contributed by one checkpoint:

$$g \equiv \frac{k_p + r_T}{k_p + r_D}. \quad (194)$$

Because the target is longer lived, $r_T < r_D$, and hence $0 < g < 1$. Therefore

$$\boxed{\eta_N^{\text{entry}} = \frac{a_D}{a_T} g^N.} \quad (195)$$

Every additional checkpoint multiplies the completed-entry error odds by the same number $g < 1$.

Continuous terminal-signal error (output definition 2)

Let's now consider output definition 2. If terminal complexes signal continuously, divide [equation \(191\)](#) for the two ligands:

$$\begin{aligned}\eta_N^{\text{cont}} &\equiv \frac{S_D}{S_T} \\ &= \frac{k_s(a_D R/r_D)p_D^N}{k_s(a_T R/r_T)p_T^N} \\ &= \frac{a_D}{a_T} \frac{r_T}{r_D} \left(\frac{p_D}{p_T}\right)^N \\ &= \boxed{\eta_{\text{eq}} \left(\frac{k_p + r_T}{k_p + r_D}\right)^N}.\end{aligned}\tag{196}$$

In the last line we used

$$\eta_{\text{eq}} = \frac{a_D}{a_T} \frac{r_T}{r_D}.\tag{197}$$

Thus

$$\boxed{\eta_N^{\text{cont}} = \eta_{\text{eq}} g^N}.\tag{198}$$

This is the main result for continuous terminal signaling: each driven checkpoint multiplies the equilibrium error odds by one additional survival factor g .

The two output conventions differ by

$$\boxed{\eta_N^{\text{cont}} = \eta_N^{\text{entry}} \frac{r_T}{r_D}}.\tag{199}$$

This extra r_T/r_D is not a hidden checkpoint. It arises after completion: a target terminal complex remains bound for mean time $1/r_T$, whereas a decoy terminal complex remains bound for only $1/r_D$. Continuous signaling rewards the target both for reaching the terminal state more often and for signaling longer after it arrives.

If target and decoy are the only sources, their wrong-output fractions are

$$\epsilon_N^{\text{entry}} = \frac{\eta_N^{\text{entry}}}{1 + \eta_N^{\text{entry}}}, \quad \epsilon_N^{\text{cont}} = \frac{\eta_N^{\text{cont}}}{1 + \eta_N^{\text{cont}}}.\tag{200}$$

For example, the fraction of the mean continuous signal that is target-derived is

$$f_T^{\text{cont}} \equiv \frac{S_T}{S_T + S_D} = \frac{1}{1 + \eta_N^{\text{cont}}}.\tag{201}$$

This is a ratio of mean signal contributions. It is not yet the probability that a whole cell makes the correct decision in a finite time.

Concentration has not disappeared:

$$\frac{a_D}{a_T} = \frac{k_{\text{on},D} c_D}{k_{\text{on},T} c_T}.\tag{202}$$

A sufficiently abundant decoy can still overwhelm a rare target. Proofreading suppresses decoy output by repeated lifetime tests. It does not create perfect recognition.

7.12 Two limiting regimes

Progression is very fast

Suppose $k_p \gg r_D > r_T$. Then

$$p_i = \frac{1}{1 + r_i/k_p} \simeq 1 - \frac{r_i}{k_p} \quad (203)$$

and

$$\begin{aligned} g &= \frac{1 + r_T/k_p}{1 + r_D/k_p} \\ &\simeq \left(1 + \frac{r_T}{k_p}\right) \left(1 - \frac{r_D}{k_p}\right) \\ &\simeq 1 - \frac{r_D - r_T}{k_p} \simeq 1. \end{aligned} \quad (204)$$

For a fixed, moderate N ,

$$\eta_N^{\text{cont}} \simeq \eta_{\text{eq}}. \quad (205)$$

Both ligands usually finish before dissociation has time to reveal their lifetime difference. Hence, we can conclude that a checkpoint that is passed almost instantaneously is not effective.

Progression is slow

Suppose $k_p \ll r_T < r_D$. Then

$$p_i = \frac{k_p}{k_p + r_i} \simeq \frac{k_p}{r_i}, \quad g \simeq \frac{r_T}{r_D}. \quad (206)$$

For completed entries,

$$\begin{aligned} \eta_N^{\text{entry}} &\simeq \frac{a_D}{a_T} \left(\frac{r_T}{r_D}\right)^N \\ &= \eta_{\text{eq}} \left(\frac{r_T}{r_D}\right)^{N-1}. \end{aligned} \quad (207)$$

For continuous terminal signaling,

$$\boxed{\eta_N^{\text{cont}} \simeq \frac{a_D}{a_T} \left(\frac{r_T}{r_D}\right)^{N+1}}. \quad (208)$$

We can now see where the two powers come from. Each of the N gates gives one lifetime factor. Continuous signaling gives one further factor because the terminal target state lasts longer.

Suppose, more restrictively, that target and decoy have equal concentrations and equal association-rate constants. Then $a_D/a_T = 1$ and

$$\eta_{\text{eq}} = \frac{r_T}{r_D}. \quad (209)$$

For entry or product counting, $N = 1$ reproduces the one-use equilibrium lifetime discrimination in the slow limit:

$$\eta_1^{\text{entry}} \simeq \eta_{\text{eq}}. \quad (210)$$

One additional gate, $N = 2$, gives

$$\eta_2^{\text{entry}} \simeq \left(\frac{r_T}{r_D}\right)^2 = \eta_{\text{eq}}^2. \quad (211)$$

This is the precise sense in which ideal proofreading can reuse, and thereby square, a one-step discrimination. For the continuous convention, ordinary bound-state signaling is the baseline set by $N = 0$, and one checkpoint yields:

$$\eta_1^{\text{cont}} \simeq \eta_{\text{eq}}^2. \quad (212)$$

There is no violation of equilibrium statistical mechanics: the driven proofreading network is not at equilibrium.

7.13 How many checkpoints are needed?

Suppose continuous signal is the chosen observable and the desired error odds are $\eta_* < \eta_{\text{eq}}$. From

$$\eta_N^{\text{cont}} = \eta_{\text{eq}} g^N \leq \eta_*, \quad (213)$$

we have

$$g^N \leq \frac{\eta_*}{\eta_{\text{eq}}}. \quad (214)$$

Because $0 < g < 1$, it is convenient to use $1/g > 1$:

$$\left(\frac{1}{g}\right)^N \geq \frac{\eta_{\text{eq}}}{\eta_*}. \quad (215)$$

Taking logarithms gives

$$N \geq \frac{\ln(\eta_{\text{eq}}/\eta_*)}{\ln(1/g)} = \frac{\ln(\eta_{\text{eq}}/\eta_*)}{\ln[(k_p + r_D)/(k_p + r_T)]}. \quad (216)$$

Since N counts discrete transitions, the smallest acceptable depth is

$$N_{\min} = \left\lceil \frac{\ln(\eta_{\text{eq}}/\eta_*)}{\ln[(k_p + r_D)/(k_p + r_T)]} \right\rceil. \quad (217)$$

If the specification is instead a maximum wrong-output fraction ϵ_* , first convert it to odds using $\eta_* = \epsilon_*/(1 - \epsilon_*)$.

7.14 The price of proofreading: accuracy versus throughput

Improving specificity has costs. We now ask how extra checkpoints affect the target output, the response time, and the energy use.

For either of our output conventions, adding one checkpoint multiplies the error odds by

$$\frac{\eta_{N+1}}{\eta_N} = g = \frac{k_p + r_T}{k_p + r_D} < 1. \quad (218)$$

Here η_N can mean either η_N^{entry} or η_N^{cont} as long as the same convention is used in numerator and denominator. Their baselines differ, but their per-checkpoint improvement factor is the same.

The target completion flux is

$$J_T(N) = a_T R p_T^N. \quad (219)$$

Adding one checkpoint changes it by

$$\frac{J_T(N+1)}{J_T(N)} = p_T < 1. \quad (220)$$

Thus the additional checkpoint rejects a larger fraction of decoys, but it also rejects some targets. Slowing k_p makes g smaller and strengthens discrimination, but it also makes p_T smaller and further reduces target throughput.

7.15 Time cost: failed attempts must be included

At any nonterminal state, the next event occurs at the total rate $k_p + r_i$, so the mean time until progression or dissociation is

$$\frac{1}{k_p + r_i}. \quad (221)$$

The probability that an attempt ever reaches state n is p_i^n : it must first pass n earlier races. Therefore the mean time spent in the preterminal part of one attempt is

$$\begin{aligned} d_i &= \sum_{n=0}^{N-1} P(\text{reach state } n) \langle \text{time in state } n \mid \text{state reached} \rangle \\ &= \sum_{n=0}^{N-1} p_i^n \frac{1}{k_p + r_i} \\ &= \frac{1}{k_p + r_i} \frac{1 - p_i^N}{1 - p_i}. \end{aligned} \quad (222)$$

Using $1 - p_i = r_i/(k_p + r_i)$ yields

$$d_i = \frac{1 - p_i^N}{r_i} = \frac{1 - q_i}{r_i}. \quad (223)$$

This clock stops when the attempt either dissociates and fails or enters the terminal state and succeeds. It does not include the later residence time in the terminal state.

Now consider one isolated ligand type and one receptor. Let $\mathcal{T}_i$ be the mean time from a free receptor to the first successful completion. Before one success, the receptor repeats the following steps:

1. It waits mean time $1/a_i$ for binding.
2. It spends mean time d_i in the bound attempt.
3. With probability $1 - q_i$, it fails, returns to the free state, and incurs another mean waiting time $\mathcal{T}_i$.

This first-step decomposition gives

$$\mathcal{T}_i = \frac{1}{a_i} + d_i + (1 - q_i)\mathcal{T}_i. \quad (224)$$

Move the restart term to the left:

$$\begin{aligned} q_i \mathcal{T}_i &= \frac{1}{a_i} + d_i, \\ \mathcal{T}_i &= \frac{a_i^{-1} + d_i}{q_i}. \end{aligned} \quad (225)$$

Substituting [equation \(223\)](#),

$$\mathcal{T}_i = \frac{a_i^{-1} + (1 - q_i)/r_i}{q_i}. \quad (226)$$

When q_i is small, many attempts fail before one succeeds, so the mean time to obtain a completed output can be very long.

There is a related but different conditional time. Given that a particular race ends in forward progression, its waiting-time density is

$$\begin{aligned} f(t \mid \text{progression wins}) &= \frac{k_p e^{-(k_p + r_i)t}}{p_i} \\ &= (k_p + r_i) e^{-(k_p + r_i)t}. \end{aligned} \quad (227)$$

Its mean is $1/(k_p + r_i)$. A successful attempt passes N such races, so

$$\langle T_{\text{proof}} \mid \text{success} \rangle = \frac{N}{k_p + r_i}. \quad (228)$$

Note that this selected time isn't the same as [equation \(226\)](#). The conditional time omits every failed attempt. Note that the rare decoy successes are precisely the trajectories in which the forward events happened unusually quickly.

7.16 Fluctuations over a finite measurement time

The ratios derived above compare mean output contributions. A cell observes a finite number of molecular events, and those numbers fluctuate. To see the issue in its simplest form, suppose terminal-entry events are approximately independent Poisson events during a measurement window T_m . If K_i is the number of entries caused by ligand i , then

$$K_i \sim \text{Poisson}(J_i T_m), \quad (229)$$

and hence

$$\langle K_i \rangle = J_i T_m, \quad \text{Var}(K_i) = J_i T_m. \quad (230)$$

The relative standard deviation is

$$\begin{aligned} \frac{\sqrt{\text{Var}(K_i)}}{\langle K_i \rangle} &= \frac{\sqrt{J_i T_m}}{J_i T_m} \\ &= \frac{1}{\sqrt{J_i T_m}}. \end{aligned} \quad (231)$$

More checkpoints decrease the mean decoy-to-target ratio, but they also lower J_T and can increase relative sampling noise within a fixed time.

The Poisson model is only an illustration. Receptor occupancy and reset can correlate events, and continuous terminal signals could contain random residence times as well as random entries. Hence, to calculate the probability that a cell makes a wrong decision, we must specify the measurement time, the downstream observable, its fluctuations, and the decision threshold. Mean specificity alone is not a complete measure of how reliable a decision is.

7.17 Energy cost: fuel is spent on failed attempts too

The forward arrows in [equation \(152\)](#) are effectively irreversible because we imagine that they are strongly driven. Suppose, for bookkeeping, that each forward step consumes one fuel molecule. The flux through transition m , from $C_{m-1}^{(i)}$ to $C_m^{(i)}$, is

$$\begin{aligned} F_m^{(i)} &= k_p C_{m-1}^{(i)} \\ &= k_p \frac{a_i R}{k_p + r_i} p_i^{m-1} \\ &= a_i R p_i^m, \quad 1 \leq m \leq N. \end{aligned} \quad (232)$$

The total fuel-use rate caused by ligand i is the sum of all forward fluxes:

$$\begin{aligned}
 \Phi_i &= \sum_{m=1}^N F_m^{(i)} \\
 &= a_i R \sum_{m=1}^N p_i^m \\
 &= \boxed{a_i R \frac{p_i(1 - p_i^N)}{1 - p_i}}.
 \end{aligned} \tag{233}$$

An attempt that advances through two steps and then dissociates has used two fuel molecules even though it produces no terminal entry. Divide the total target fuel flux by the successful target-entry flux:

$$\begin{aligned}
 \frac{\Phi_T}{J_T} &= \frac{a_T R \sum_{m=1}^N p_T^m}{a_T R p_T^N} \\
 &= \sum_{m=1}^N p_T^{m-N} \\
 &= \boxed{\sum_{j=0}^{N-1} p_T^{-j}}.
 \end{aligned} \tag{234}$$

If $p_T \simeq 1$, this is approximately N fuel reactions per completed target. If p_T is small, failed attempts make the cost per success much larger. For example, if $p_T = 1/2$ and $N = 6$, then

$$\frac{\Phi_T}{J_T} = 1 + 2 + 4 + 8 + 16 + 32 = 63 \tag{235}$$

the model uses 63 fuel-consuming forward reactions per completed target entry.

If each fuel reaction is driven by chemical-potential drop $\Delta\mu$, the corresponding chemical power is

$$\dot{W}_{\text{chem}} = \Delta\mu \sum_i \Phi_i. \tag{236}$$

A complete thermodynamic model must include reverse reactions, finite chemical affinities, and any bypass pathways. These processes limit the accuracy attainable at finite dissipation [10, 11].

7.18 What makes this proofreading rather than an ordinary cascade?

A multistep phosphorylation cascade is not automatically a proofreading network. The derivation of $q_i = p_i^N$ requires:

1. Intermediate states are traversed while the ligand remains bound.
2. Progression competes with ligand dissociation.
3. Progress is lost when that ligand dissociates.
4. Driven forward reactions prevent ordinary equilibration from erasing the benefit of repeated selection.

The reset condition is especially important. If downstream modifications survive after receptor unbinding, several short encounters can accumulate the same progress as one long encounter. Completion would then no longer require one uninterrupted bound lifetime, and the simple product p_i^N would fail.

An ordinary MAPK pathway may relay, amplify, delay, or low-pass filter a signal without tying downstream reset to the survival of the original receptor–ligand complex. Such a cascade can be biologically valuable without performing kinetic proofreading. A real pathway may combine both functions, but relay and proofreading are mathematically distinct.

7.19 What the minimal model leaves out

We obtained this power law because of the assumptions above. Real signaling pathways may violate these assumptions.

- Forward rates may differ between states. If state n has rate $k_{p,n}$ and dissociation still has rate r_i , then

$$q_i = \prod_{n=0}^{N-1} \frac{k_{p,n}}{k_{p,n} + r_i}. \quad (237)$$

- Target and decoy may differ in forward rates as well as off-rates. The equal- k_p model lets us study lifetime discrimination by itself, but real pathways do not need to have equal forward rates.
- Dissociation rates may change after modification.
- Reset may be slow or incomplete, and rapid rebinding may allow one ligand to recover its old intermediate state.
- Reverse reactions and finite chemical driving weaken the ideal irreversible limit.
- Receptors may cooperate, saturate downstream enzymes, or participate in feedback loops.
- Ligand concentrations and receptor occupancy may change before steady state is reached.

Each extension changes the calculation. In each case, we must specify the states, the rates, what resets after unbinding, which steps use chemical energy, and what output we measure.

Kinetic proofreading in one statement

In the ideal model we considered here, each checkpoint is a race between forward progression and ligand dissociation:

$$p_i = \frac{k_p}{k_p + r_i}, \quad q_i = p_i^N.$$

Because a short-lived decoy has $p_D < p_T$, repeating the race multiplies discrimination factors. For continuous terminal signaling,

$$\eta_N^{\text{cont}} = \eta_{\text{eq}} \left(\frac{k_p + r_T}{k_p + r_D} \right)^N.$$

Complete reset ties every checkpoint to one uninterrupted binding event. Chemical driving keeps the sequence directional. This increased specificity has costs. Fewer target attempts finish, the cell waits longer for successful events, finite-time fluctuations become larger, and the forward steps use fuel.

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